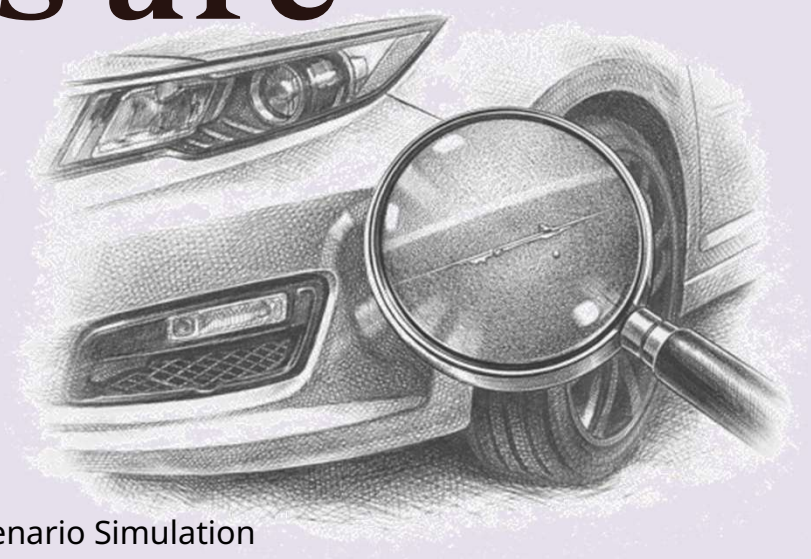
A probabilistic approach to quality inspection
What if surface defects followed patterns instead of chance?

Visual defects are not random.



Probability Modeling | Risk Estimation| Scenario Simulation

The problem with visual defects

> **Visual defects are one of the most expensive and least understood sources of quality loss.**
>
> They increase scrap and rework, slow down operations, and introduce subjectivity at inspection — often without a clear root cause.

Defects don’t happen. They emerge.

From process to risk

> **Visual defects are detected at inspection but created upstream.**
>
> What looks like a sudden failure at the end of the line is usually the result of accumulated process conditions across injection, storage, handling, and painting.

Injection → Storage → Handling → Painting → Inspection

If defects are not random..., where do they really come from?

The data behind the problem

- The dataset represents typical manufacturing and paint shop conditions observed in plastic automotive components.
- It reflects information collected across multiple stages of the process, not a single inspection point.
- The data captures process conditions, environmental factors, and handling intensity that influence cosmetic risk.

Data preparation required aligning information from:

1. Injection molding parameters (material, temperature, cooling conditions)
2. Pre-paint storage and handling conditions (time, movements, protection)
3. Paint shop parameters (viscosity, film thickness, humidity)
4. Final visual inspection outcomes (pass / fail)
5. Building a usable dataset was not a trivial step — it was the foundation of the analysis.

Defects are visible at inspection, but measurable much earlier.

The model: Why Naive Bayes

> **Visual defect inspection is a screening problem, not a prediction contest.**
>
> Naive Bayes is especially effective when multiple weak signals combine to form a risk — even when those signals come from different stages of the process and are only approximately independent.

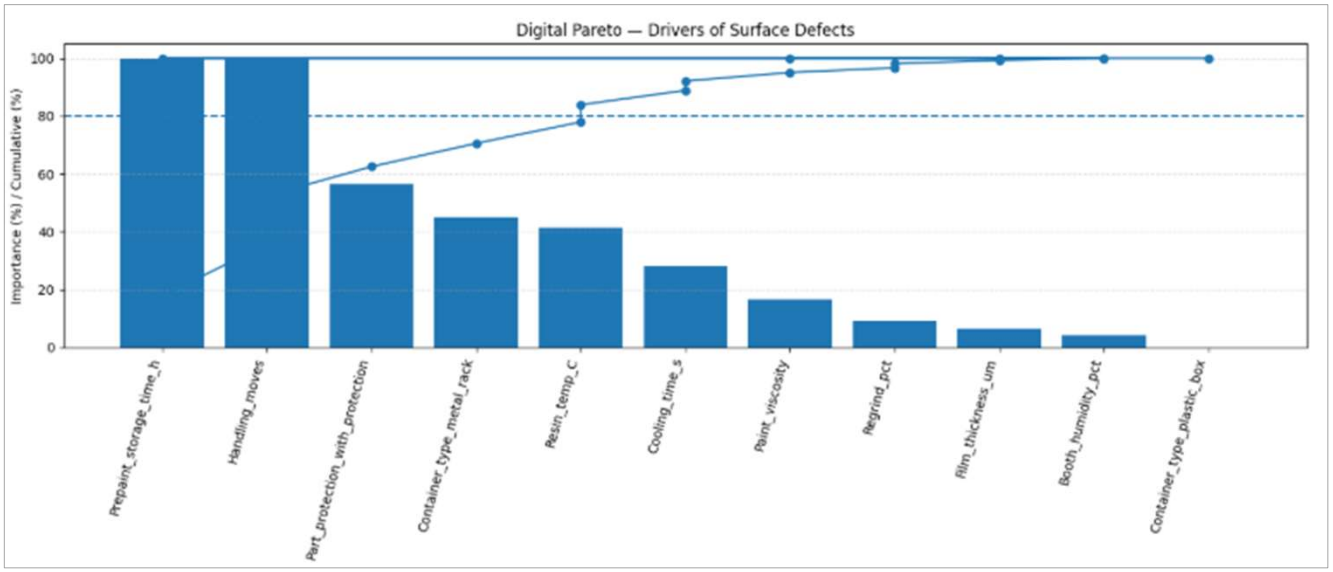
- Probabilistic by design — outputs risk, not certainty
- Fast and lightweight — suitable for near real-time screening
- Interpretable — easy to explain to non-technical teams
- Robust when signals are partially independent

This model does not decide. It estimates risk.

Key Drivers of Visual Defect Risk

Not all process variables contribute equally to visual defects.

A small number of conditions explain most of the observed cosmetic risk.



Prepaint_storage_time_h	100.00
Handling_moves	100.00
Handling_moves	94.10
Part_protection_with_protection	56.59
Container_type_metal_rack	45.11
Resin_temp_C	41.55
Resin_temp_C	33.31
Cooling_time_s	28.01
Cooling_time_s	18.09
Paint_viscosity	16.66

Most visual defects are driven by a small set of upstream conditions.

What-If Scenarios — From Conditions to Risk

> The same part can carry very different levels of visual risk depending on process conditions.
>
> The simulator allows us to explore scenarios without changing the line.

01

```python
# Good conditions
prob, decision = simulate_surface_defect_risk(
    nb_model,
    Regrind_pct=5,
    Resin_temp_C=235,
    Cooling_time_s=20,
    Paint_viscosity=25,
    Film_thickness_um=30,
    Booth_humidity_pct=50,
    Prepaint_storage_time_h=1,
    Handling_moves=2,
    Container_type="plastic_box",
    Part_protection="with_protection",
    threshold=0.50
)

print(f"Defect probability: {prob:.3f} | Decision: {decision}")
```

```
Defect probability: 0.046 | Decision: PASS (Low/Moderate Risk)
```

02

```python
# Bad handling + bad enviroment :(
prob, decision = simulate_surface_defect_risk(
    nb_model,
    Regrind_pct=28,
    Resin_temp_C=248,
    Cooling_time_s=13,
    Paint_viscosity=21,
    Film_thickness_um=38,
    Booth_humidity_pct=68,
    Prepaint_storage_time_h=18,
    Handling_moves=6,
    Container_type="metal_rack",
    Part_protection="no_protection",
    threshold=0.50
)

print(f"Defect probability: {prob:.3f} | Decision: {decision}")
```

```
Defect probability: 0.977 | Decision: FAIL (High Risk)
```

So what do we learn from all this?

Final reflection

This project is not about predicting defects with certainty.
It is about making cosmetic risk visible.
Visual defects are rarely the result of a single mistake.

Naive Bayes allows us to translate weak signals from injection, storage, handling, and painting into a probability that teams can understand and discuss.

It is a conversation:
What conditions are we tolerating today?
Where is risk quietly accumulating?
Which levers are worth improving?

Delays are not random events. They are the result of interacting processes.

Take it further

Feel free to download the full project:
the dataset, the notebook, and the simulator.

Change the data.
Break the code.
Ask different questions.

This model is just one snapshot in an ongoing process.
And that process never truly stops.

Because improvement has no finish line.